

Metoclopramide: Drug information - UpToDate

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For additional information see "Metoclopramide: Patient drug information" and "Metoclopramide: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Tardive dyskinesia:

Metoclopramide can cause tardive dyskinesia, a potentially irreversible serious movement disorder. In patients treated with metoclopramide, the risk of developing tardive dyskinesia increases with duration of treatment and total cumulative dose. Metoclopramide is contraindicated in patients with a history of tardive dyskinesia. Use metoclopramide for the shortest duration of treatment and periodically reassess the need for continued treatment. Immediately discontinue metoclopramide in patients who develop signs or symptoms of tardive dyskinesia. In patients with symptomatic, documented gastroesophageal reflux, the maximum duration of metoclopramide treatment is 12 weeks. Avoid a total duration of treatment with metoclopramide products for longer than 12 weeks. If longer-term use is unavoidable, routinely monitor for signs and symptoms of tardive dyskinesia.

Brand Names: US

- Gimoti;
- Reglan

Brand Names: Canada

- MAR-Metoclopramide;
- Metonia [DSC];
- PMS-Metoclopramide;
- PMS-Metoclopramide HCl;
- PRZ-Metoclopramide

Pharmacologic Category

- Antiemetic;
- Dopamine Antagonist;
- Gastrointestinal Agent, Prokinetic;
- Serotonin 5-HT₄ Receptor Agonist

Dosing: Adult

Dosage guidance:

Safety: To decrease the risk of tardive dyskinesia (TD), use for the shortest duration of treatment and periodically reassess the need for continued treatment; do **not** use continuously for ≥ 12 weeks. If longer-term use is unavoidable, routinely monitor for signs and symptoms of TD. Discontinue immediately if the diagnosis of TD is made; consider the use of alternative agents.

Dosing: In Canada, product information limits the total daily dose to 0.5 mg/kg except when used for prophylaxis of delayed emesis associated with cisplatin chemotherapy.

Aspiration prophylaxis in patients undergoing anesthesia

Aspiration prophylaxis in patients undergoing anesthesia (off-label use):

Note: May be considered in patients at high risk for aspiration (Ref):

IV: 10 mg administered over 1 to 2 minutes as a single dose ~30 to 60 minutes prior to induction of anesthesia; usually given with nonparticulate antacid(s) (eg, oral sodium citrate, citric acid) and/or an H_2 receptor antagonist (Ref).

Bowel obstruction, malignant inoperable

Bowel obstruction (partial), malignant inoperable (off-label use):

Note: Do **not** use in patients with confirmed or suspected complete mechanical obstruction (Ref). Often given in combination regimen (eg, with an antisecretory agent, corticosteroid, and/or anticholinergic) (Ref). Avoid metoclopramide use if colic or abdominal pain are present (Ref).

Oral, IV, SUBQ: 10 mg every 4 to 6 hours (Ref); if insufficient relief with intermittent dosing, may switch to an IV or SUBQ continuous infusion.

Continuous IV or SUBQ infusion: Initial: 1.25 to 1.75 mg/hour; titrate as needed and tolerated up to 5 mg/hour (Ref).

Chemotherapy-induced nausea and vomiting, prevention

Chemotherapy-induced nausea and vomiting, prevention (alternative agent):

Note: Not for routine use; may be considered in select patients (eg, patients refractory to first-line agents or patients with a history of nausea/vomiting following low-emetic-risk regimens). Other agents are safer and more effective (Ref).

Low-emetic-risk IV chemotherapy:

Oral: Dosing regimen based on expert opinion: 10 mg before chemotherapy **or** 10 mg every 6 hours as needed (Ref).

Prophylaxis of delayed emesis (high-emetic-risk chemotherapy regimen) (alternative agent) (off-label use):

Note: Routine use is **not** recommended for prophylaxis of delayed emesis (alternative agents recommended); however, potential alternative to neurokinin 1 (NK_1) receptor antagonist (eg, aprepitant, fosaprepitant) in patients receiving cisplatin.

Oral: 10 to 20 mg 4 times daily on postchemotherapy days 2 through 4; given in combination with dexamethasone (Ref).

Dyspepsia, functional

Dyspepsia, functional (refractory) (off-label use):

Note: For patients unresponsive to first-line therapies (eg, proton pump inhibitors, *Helicobacter pylori* eradication, and/or tricyclic antidepressants) (Ref); some experts recommend limiting duration of metoclopramide therapy to 4 weeks (Ref):

Oral: 5 to 10 mg 3 to 4 times daily administered prior to meals and at bedtime (Ref). A lower dose of 2 mg 3 times daily (using oral liquid formulation) may provide sufficient response (Ref).

Gastroparesis, diabetic and nondiabetic

Gastroparesis, diabetic (labeled use) and nondiabetic (off-label use):

Note: For patients who have had insufficient response to appropriate initial interventions (eg, dietary therapy, discontinuing drugs that impair GI motility, improved glycemic control in diabetic gastroparesis) (Ref). With diabetic gastroparesis, some experts suggest limiting use to severe refractory cases (Ref). Initiating treatment with an oral liquid formulation is preferred and may improve absorption (Ref).

Nasal: One spray (15 mg) in one nostril 4 times daily (30 minutes prior to each meal and at bedtime) for 2 to 8 weeks depending on symptomatic response; maximum: 4 sprays/day (60 mg/day).

Missed/incomplete doses: If a dose is missed or it is uncertain whether the spray entered the nose, do not repeat or make up dose; administer dose at the next scheduled time.

Oral (preferred), IM, IV, SUBQ: 5 mg 2 or 3 times daily administered 30 minutes prior to meals; may gradually increase up to 10 mg 30 minutes prior to meals and at bedtime if needed based on response. Titrate to the lowest effective dose; maximum: 40 mg/day in 4 divided doses (Ref). If parenteral administration is indicated, some experts do not exceed 30 mg/day in 3 divided doses (Ref). In gastroparesis associated with anorexia nervosa, a lower oral dose of 2.5 mg administered prior to each meal and at bedtime may provide sufficient symptom relief according to some experts (Ref).

Duration: In chronic therapy, limit course to ≤ 12 weeks. Consider a “drug holiday” or dose reduction (eg, 5 mg twice daily before two main meals of the day) for ~ 2 weeks whenever clinically feasible or at least every 12 weeks (whichever is shorter) to evaluate efficacy and necessity of continued treatment (Ref).

Hiccups

Hiccups (alternative agent) (off-label use):

Note: Limited data available. Treatment generally continues for up to 5 to 10 days. May discontinue treatment the day after hiccups subside; if hiccups recur, longer treatment duration may be needed (eg, in palliative care) (Ref).

IV: 5 to 10 mg every 8 hours (Ref).

Oral: 10 mg every 6 to 8 hours (Ref).

Medication-overuse headache or intractable migraine

Medication-overuse headache or intractable migraine (status migrainosus) (adjunctive agent) (off-label use):

Note: Use as part of an appropriate combination regimen or as adjunctive therapy for prevention of nausea and vomiting in patients receiving dihydroergotamine (Ref). Premedication with diphenhydramine is suggested to prevent akathisia and other acute dystonic reactions (Ref). Practice varies by institutional protocol; example regimen below.

IV: Initial: 10 mg administered immediately prior to dihydroergotamine; repeat as needed every 8 hours, either prior to each intermittent dihydroergotamine dose or during continuous IV administration of dihydroergotamine (Ref). Doses of metoclopramide up to 20 mg have been used in patients experiencing severe nausea (Ref).

Migraine, severe, acute treatment

Migraine, severe, acute treatment (emergency setting) (off-label use):

Note: Use as monotherapy or as part of an appropriate combination regimen (Ref). IV administration may be more efficacious (Ref). Premedication with diphenhydramine is suggested to prevent akathisia and other acute dystonic reactions (Ref).

IV (preferred), IM, Oral, SUBQ: 10 mg as a single dose; for migraine with severe nausea and vomiting, some experts increase the dose to 20 mg (Ref).

Nausea and/or vomiting

Nausea and/or vomiting:

Undifferentiated or due to a variety of medical conditions associated with acute self-limiting nausea/vomiting (alternative agent) (off-label use): Limited data available:

IV: 10 mg or 20 mg as a single dose (Ref); avoid rapid IV administration of doses >10 mg.

Oral: 10 mg as a single dose; may repeat after 4 to 6 hours if needed.

Advanced cancer-associated (off-label use):

Note: Limited data available; dosing recommendations based primarily on expert opinion:

Empiric therapy in patients without an identifiable etiology:

Oral, IV, SUBQ: 10 mg every 6 hours (Ref).

Malignancy-associated gastroparesis:

Oral, IV, SUBQ: 5 to 10 mg every 6 hours **or** 3 to 4 times daily before meals and at night (Ref).

Palliative care setting:

Oral, IV, SUBQ: Initial: 5 to 10 mg every 4 to 6 hours (Ref). For severe symptoms, some experts titrate up to 100 mg/day (in divided doses) if needed (Ref). If insufficient relief with intermittent dosing, may switch to an IV or SUBQ continuous infusion.

Continuous IV or SUBQ infusion: Initial: 1.25 to 1.75 mg/hour; titrate as needed and tolerated up to 5 mg/hour (Ref).

Nasal: One spray (15 mg) in one nostril 4 times daily (30 minutes prior to each meal and at bedtime) for 2 to 8 weeks depending on symptomatic response; maximum: 4 sprays (60 mg)/day (Ref).

Missed/incomplete doses: If a dose is missed or it is uncertain whether the spray entered the nose, do not repeat or make up dose; take dose at the next scheduled time.

Pregnancy-associated, severe or refractory (off-label use):

Note: May be considered as add-on or alternative therapy for nausea and vomiting when symptoms persist following initial pharmacologic therapy and IV hydration if hypovolemic (Ref).

Oral, IV, IM: 5 to 10 mg every 6 to 8 hours, added to current treatment regimen or as alternative therapy; if feasible, give 30 minutes before meals and bedtime (Ref).

Vertigo-associated (alternative agent) (off-label use):

Oral, IV: 5 to 10 mg every 6 hours as needed (Ref).

Postoperative nausea and vomiting, prophylaxis

Postoperative nausea and vomiting, prophylaxis (alternative agent):

Note: Available data to support use in this condition are limited (Ref). May use when dopamine antagonists are indicated but other preferred agents (eg, droperidol) cannot be used (Ref).

IM, IV (off-label route): 10 mg administered near the end of surgery (Ref). **Note:** Higher doses (eg, 20 to 50 mg) may be more efficacious; however, some experts avoid use of these doses due to increased risk of adverse reactions (eg, hypotension, tachycardia, extrapyramidal effects) (Ref).

Radiation therapy-induced nausea and vomiting

Radiation therapy-induced nausea and vomiting (rescue therapy) (alternative agent) (off-label use):***Low-emetic-risk radiation therapy (head and neck, thorax, or pelvis):***

Oral, IV: 5 to 20 mg if needed after each radiation treatment and repeated every 6 to 8 hours; depending on symptom severity and remaining duration of radiation therapy, patients can receive subsequent rescue therapy as needed or begin prophylactic therapy (Ref).

Minimal-emetic-risk radiation therapy (extremities, breast):

Oral, IV: 5 to 20 mg if needed after each radiation treatment (Ref).

Note: Avoid rapid IV administration of doses >10 mg.

Small bowel intubation, postpyloric feeding tube placement

Small bowel intubation, postpyloric feeding tube placement:

Note: Available data to support use in this condition are limited (Ref).

IV: 10 to 20 mg as a single dose administered 10 to 15 minutes prior to tube insertion (Ref).

Tension-type headache, acute

Tension-type headache, acute (emergency setting) (alternative agent) (off-label use):

Note: Limited data available.

IV: 10 mg as a single dose (Ref) **or** 20 mg as a single dose; premedicate with IV diphenhydramine to prevent akathisia and other acute dystonic reactions (Ref). **Note:** Avoid rapid IV administration of metoclopramide doses >10 mg.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function (Ref):

Injection, Oral:

CrCl >60 mL/minute: No dosage adjustment necessary.

CrCl >10 to 60 mL/minute: Administer ~50% of usual total daily dose.

CrCl ≤10 mL/minute: Administer ~33% (or less) of usual total daily dose.

Nasal:

CrCl >60 mL/minute: No dosage adjustment necessary.

CrCl ≤60 mL/minute: Use not recommended; dose cannot be easily adjusted.

Hemodialysis, intermittent (thrice weekly):

Injection, Oral: Not significantly dialyzed (Ref): Administer ~33% (or less) of usual total daily dose (Ref).

Nasal: Use not recommended; dose cannot be easily adjusted.

Peritoneal dialysis:

Injection, Oral: Not significantly dialyzed (Ref): Administer ~33% (or less) of usual total daily dose (Ref).

Nasal: Use not recommended; dose cannot be easily adjusted.

CRRT :

Injection, Oral: Not likely to be substantially removed due to large volume of distribution: Administer ~50% of usual total daily dose (Ref).

Nasal: Use not recommended; dose cannot be easily adjusted.

PIRRT (eg, sustained, low-efficiency diafiltration):

Injection, Oral: Not likely to be substantially removed due to large volume of distribution: Administer ~50% of usual total daily dose (Ref).

Nasal: Use not recommended; dose cannot be easily adjusted.

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matthew Arango, PharmD, BCOP; Justin Floyd, DO, FACOI; Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Metoclopramide elimination is significantly impaired (eg, clearance decreased by ~50% and half-life prolonged by ~2-fold) in the setting of liver cirrhosis (Ref).

Liver impairment prior to treatment initiation:

Initial or dose titration in patients with preexisting liver cirrhosis:

Child-Turcotte-Pugh class A: Injection, Nasal, Oral: No dosage adjustment necessary (Ref).

Child-Turcotte-Pugh class B and C:

Injection, Oral: Administer 50% of the usual indication specific dose and consider extending the dosing interval (eg, if normal dosing interval is every 6 hours then, extend to every 8 hours) (Ref).

Nasal: Note: Manufacturer's labeling for the nasal formulation states use not recommended in Child-Turcotte-Pugh class B and C; however, based on the known pharmacokinetics of the nasal formulation, use in these patients is likely safe (Ref).

No dosage adjustment necessary (Ref).

Dosing: Older Adult

Note: Avoid use, except for gastroparesis. Limit duration to ≤ 12 weeks, with rare exceptions (Ref).

Nasal: Not recommended as initial therapy. Patients stable on 10 mg 4 times daily of other formulations may be switched to 1 spray (15 mg) 4 times daily for 2 to 8 weeks depending on symptomatic response. Refer to adult dosing.

Oral/Injection: Initial: Dose at the lower end of the recommended range (may require only 5 mg/dose) and use the lowest effective dose. Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Metoclopramide: Pediatric drug information")

Chemotherapy-induced nausea and vomiting, prevention, alternative agent

Chemotherapy-induced nausea and vomiting (CINV), prevention, alternative agent: Limited data available; optimal dose not established; efficacy results variable:

Note: Guidelines do not recommend metoclopramide for prevention of CINV due to lack of proven efficacy and safety (Ref). Metoclopramide may be considered for breakthrough CINV in patients receiving highly emetogenic chemotherapy who cannot receive olanzapine and for the prevention of refractory CINV after assessment of risk versus benefit (Ref). Doses required to prevent CINV are high and should not be used first-line due to increased risk of extrapyramidal symptoms (acute dystonic reactions); concurrent administration of diphenhydramine or benztropine is recommended to prevent drug-induced adverse effects (Ref).

Children and Adolescents: IV: 0.5 to 2 mg/kg/dose prior to chemotherapy; may repeat up to 4 times daily; usual adult dose: 10 mg/dose (Ref); although more frequent dosing has been described (every 2 to 4 hours), adverse events were more common (Ref). To limit the risk of extrapyramidal symptoms, more restrictive dosing of 0.1 to 0.15 mg/kg/dose every 6 hours for up to 3 doses (maximum daily dose: 0.5 mg/kg/day) is approved in Canada and Europe (Ref).

Gastroesophageal reflux, treatment

Gastroesophageal reflux, treatment: Note: Routine use is not recommended; reserve use as a last resort after all other therapies have failed and following consultation with a GI specialist (Ref). Limited data available; efficacy results variable:

Infants, Children, and Adolescents: Oral: 0.1 to 0.2 mg/kg/dose every 6 to 8 hours; maximum dose: 10 mg/dose (Ref).

Postoperative nausea and vomiting, prevention

Postoperative nausea and vomiting (PONV), prevention: Limited data available; optimal dose not established; efficacy results variable:

Note: Guidelines do not include metoclopramide as a therapeutic option for pediatric patients; other agents have shown superior efficacy with fewer adverse reactions(Ref).

Children and Adolescents: IV: 0.1 to 0.25 mg/kg/dose as a single dose administered after induction or on arrival to postanesthesia care unit; maximum dose: 10 mg/dose (Ref). **Note:** Although higher doses have been reported for PONV, use may be associated with an increased risk of adverse effects (Ref).

Postpyloric feeding tube placement

Postpyloric feeding tube placement (small bowel intubation): Note: Use in patients who have failed conventional measures.

Infants and Children <6 years: IV: 0.1 mg/kg as a single dose.

Children ≥6 years and Adolescents ≤14 years: IV: 2.5 to 5 mg as a single dose.

Adolescents ≥15 years: IV: 10 mg as a single dose.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents: The following adjustments have been recommended (Ref). **Note:** Renally adjusted dose recommendations are based on IV or oral doses of 0.1 to 0.2 mg/kg/dose every 6 to 8 hours.

GFR >50 mL/minute/1.73 m²: No adjustment necessary.

GFR 30 to 50 mL/minute/1.73 m²: Administer 75% of dose.

GFR 10 to 29 mL/minute/1.73 m²: Administer 50% of dose.

GFR <10 mL/minute/1.73 m²: Administer 25% of dose.

Intermittent hemodialysis: Administer 25% of dose.

Peritoneal dialysis (PD): Administer 25% of dose.

Continuous renal replacement therapy (CRRT): Administer 75% of dose.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; however, metoclopramide has been used safely in adult patients with advanced liver disease with normal renal function.

Adverse Reactions (Significant): Considerations

CNS depression

Drowsiness, fatigue, lassitude, dizziness, and confusion have occurred with therapeutic and supratherapeutic dosing. These effects are self-limiting and usually resolve within 24 hours of drug discontinuation.

Mechanism: Dose-related; related to the pharmacologic action (ie, dopamine receptor antagonism).

Onset: Rapid; typically occurs within the first 5 days of use (median: 1 day) (Ref).

Risk factors:

- Age >65 years (Ref)
- Concurrent use of CNS depressants
- Kidney impairment

Extrapyramidal symptoms

Drug-induced extrapyramidal reaction may occur with metoclopramide. Acute **dystonic reactions** have occurred, such as involuntary limb movements, facial grimacing, torticollis, oculogyric crisis, rhythmic tongue protrusion, bulbar type of speech, trismus, or dystonic reactions resembling tetanus. Stridor and dyspnea occurred rarely. Acute dystonic reactions are generally reversible with drug discontinuation and may require intervention. **Akathisia** (motor **restlessness**), manifesting as feelings of anxiety, agitation, insomnia, inability to sit still, pacing, and foot tapping has occurred. Akathisia is generally reversible with drug discontinuation or dose reduction. **Parkinsonism** symptoms, such as bradykinesia, tremor, cogwheel rigidity, and mask-like facies, have occurred after longer courses of therapy. Parkinsonian symptoms generally resolve within 2 to 3 months after drug discontinuation. **Tardive dyskinesia** (TD), a syndrome of disfiguring involuntary movements of the face, tongue, trunk, and/or extremities, has occurred. These effects are potentially irreversible; however, symptoms may decrease or resolve after drug discontinuation in some patients (Ref). Earlier detection of TD with immediate drug discontinuation may correspond to a higher likelihood of reversibility.

Mechanism: Dose- and time-related; related to the pharmacologic action (ie, dopamine receptor antagonism) (Ref).

Onset:

- *Acute dystonia:* Rapid; usually occurs within the first 24 to 48 hours of therapy initiation
- *Akathisia:* Rapid; usually occurs within the first few hours of therapy initiation, but can occur up to 48 hours after drug administration (Ref)
- *Parkinsonism:* Delayed; typically occurs within the first 6 months of therapy, but may present after longer periods
- *Tardive dyskinesia:* Delayed; typically occurs after the first 12 weeks of therapy

Risk factors:

General:

- Higher than recommended doses (Ref)
- Longer durations of therapy (>12 weeks)
- Age ≥60 years (Ref) and pediatric patients
- Females
- Kidney impairment
- Diabetes mellitus
- Concurrent use of other drugs that can cause extrapyramidal symptoms (eg, antipsychotics)
- Concurrent use of strong CYP2D6 inhibitors (eg, fluoxetine, paroxetine, bupropion) (Ref)
- CYP2D6 poor metabolizers

Specific:

- Acute dystonia: Age <30 years
- Akathisia: Rapid intravenous administration (eg, IV bolus over 2 minutes) (Ref)

- Parkinsonism: History of Parkinson disease, concurrent use of dopamine agonists and drugs that increase dopamine

Hyperprolactinemia

Galactorrhea not associated with childbirth, amenorrhea, gynecomastia, and erectile dysfunction have occurred. These effects are generally reversible after drug discontinuation. Although drug-induced hyperprolactinemia is usually associated with prolactin levels of 25 to 100 mcg/L, metoclopramide use may lead to prolactin levels higher than 200 mcg/L (Ref). While many reports of symptomatic hyperprolactinemia have corresponded with elevated prolactin levels, symptoms despite normal prolactin levels have also been described (Ref).

Mechanism: Related to the pharmacologic action (ie, dopamine receptor antagonism).

Onset: Varied; may occur within the first week of therapy up to 4 months after therapy initiation (Ref).

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Gastrointestinal: Dysgeusia (nasal spray: 15%)

Nervous system: Drowsiness (~10% to 70%; dose related), dystonic reaction ($\leq$ 25%; dose and age related)

1% to 10%: Nervous system: Fatigue (~10%; dose related), lassitude (~10%; dose related), restlessness (~10%; dose related)

Frequency not always defined:

Cardiovascular: Atrioventricular block, bradycardia, flushing (following high IV doses), hypertension (severe) (Ref), hypotension (rare) (Ref), supraventricular tachycardia

Dermatologic: Skin rash, urticaria

Endocrine & metabolic: Amenorrhea, endocrine disease (elevation of aldosterone), fluid retention, galactorrhea not associated with childbirth, hyperprolactinemia, porphyria

Gastrointestinal: Change in bowel habits, diarrhea, nausea

Genitourinary: Urinary frequency, urinary incontinence

Hematologic & oncologic: Leukopenia, methemoglobinemia, neutropenia

Hypersensitivity: Tongue edema

Nervous system: Confusion, dizziness, hallucination, headache, insomnia, seizure

Neuromuscular & skeletal: Laryngospasm (rare)

Ophthalmic: Visual disturbance

Respiratory: Bronchospasm, laryngeal edema

Postmarketing:

Cardiovascular: Cardiac failure (Ref), sinoatrial arrest (usually with rapid IV administration or higher doses) (Ref)

Endocrine & metabolic: Gynecomastia (Ref)

Hematologic & oncologic: Agranulocytosis (Ref), sulfhemoglobinemia (Ref)

Hypersensitivity: Angioedema (Ref)

Nervous system: Akathisia (rare: <1%) (Ref), depression (Ref), drug-induced extrapyramidal reaction (rare: <1%) (Ref), neuroleptic malignant syndrome (Ref), parkinsonism (rare: <1%) (Ref), suicidal ideation (Ref), tardive dyskinesia (total cumulative dose and duration of treatment related; rare: <1%) (Ref)

Contraindications

Hypersensitivity (eg, laryngeal and glossal angioedema, bronchospasm) to metoclopramide or any component of the formulation; situations where stimulation of gastrointestinal (GI) motility may be dangerous, including mechanical GI obstruction, perforation, or hemorrhage (except when used prior to endoscopy for evaluation of acute upper GI bleeding [Barkun 2010]); pheochromocytoma or other catecholamine-releasing paragangliomas; seizure disorder (eg, epilepsy); history of tardive dyskinesia or dystonic reaction to metoclopramide; concomitant use with other agents likely to increase extrapyramidal reactions.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Canadian labeling: Additional contraindications (not in the US labeling): Infants <1 year of age.

Warnings/Precautions

Concerns related to adverse effects:

- Hypertension: May elevate BP; avoid use in patients with hypertension (IV administration was shown to release catecholamines). There are reports of hypertensive crises in patients with undiagnosed pheochromocytoma (Leonard 2018).

Disease-related concerns:

- Cancer (eg, breast cancer): The clinical significance of hyperprolactinemia in patients with breast cancer or other prolactin-dependent tumors is unknown.
- Edematous conditions: Use with caution in patients who are at risk of fluid overload (HF, cirrhosis). Metoclopramide causes a transient increase in serum aldosterone and increases the risk for fluid retention and volume overload.
- Liver impairment: Use caution in patients with moderate to severe hepatic impairment; risk of adverse reactions may be increased due to increased systemic exposure.
- Nicotinamide adenine dinucleotide hydrogen (NADH)-cytochrome b5 reductase deficiency: Patients with NADH-cytochrome b5 reductase deficiency are at increased risk of methemoglobinemia and/or sulfhemoglobinemia.
- Parkinson disease: Symptoms of Parkinson disease may be exacerbated; avoid use in patients with Parkinson disease and other patients being treated with antiparkinsonian drugs.
- Renal impairment: Use with caution in patients with moderate to severe renal impairment; risk of adverse reactions may be increased due to increased systemic exposure. Use of nasal formulation is not recommended since dose is not easily adjusted to reduce exposure.
- Surgical anastomosis/closure: Use with caution following surgical anastomosis/closure; promotility agents may theoretically increase pressure in suture lines.

Special populations:

- Bariatric surgery: Presurgical assessment of the indication for use, symptoms, and goals of therapy should be documented to enable postsurgical assessment. Monitor for continued efficacy after bariatric surgery and consider switching to an alternative formulation or medication if symptoms worsen.
- CYP2D6 poor metabolizers: Elimination of metoclopramide may be slowed possibly increasing the risk of dystonic and other adverse reactions; dosage reduction recommended.
- Pediatric: Not recommended for use (nasal/oral formulations) due to increased risk of tardive dyskinesia and other extrapyramidal symptoms as well as the risk of methemoglobinemia in neonates

(neonates may have decreased levels of NADH-cytochrome b5 reductase, which increases the risk of methemoglobinemia).

Dosage form specific issues:

- Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Zar 2007). See manufacturer's labeling.

Warnings: Additional Pediatric Considerations

Metoclopramide is not recommended for the management of gastroesophageal reflux in pediatric patients due to increased risk of adverse events and lack of data showing efficacy (AAP [Eichenwald 2018]; NASPGHAN/ESPGHAN [Rosen 2018]); other therapeutic agents should be considered.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Generic: 5 mg/mL (2 mL)

Solution, Injection [preservative free]:

Generic: 5 mg/mL (2 mL)

Solution, Nasal:

Gimoti: 15 mg/actuation (9.8 mL) [contains benzalkonium chloride, edetate (edta) disodium dihydrate]

Solution, Oral:

Generic: 5 mg/5 mL (10 mL [DSC], 473 mL); 10 mg/10 mL (10 mL, 473 mL [DSC])

Tablet, Oral:

Reglan: 5 mg [contains fd&c blue #1 (brill blue) aluminum lake, quinoline (d&c yellow #10) aluminum lake]

Reglan: 10 mg [dye free]

Generic: 5 mg, 10 mg

Tablet Disintegrating, Oral:

Generic: 5 mg [DSC]

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Gimoti Nasal)

15 mg/ACT (per mL): \$312.37

Solution (Metoclopramide HCl Injection)

5 mg/mL (per mL): \$0.71 - \$5.03

Solution (Metoclopramide HCl Oral)

5 mg/5 mL (per mL): \$0.21 - \$1.27

Tablets (Metoclopramide HCl Oral)

5 mg (per each): \$0.32 - \$0.90

10 mg (per each): \$0.28 - \$0.96

Tablets (Reglan Oral)

5 mg (per each): \$4.68

10 mg (per each): \$4.68

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Generic: 5 mg/mL (2 mL, 10 mL, 30 mL)

Solution, Oral:

Generic: 5 mg/5 mL (250 mL, 500 mL)

Tablet, Oral:

Metonia: 10 mg [DSC]

Generic: 5 mg, 10 mg

Administration: Adult

Injection: May be given IM, direct IV push, short infusion (at least 15 minutes), or continuous infusion; lower doses (≤ 10 mg) of metoclopramide can be given IV push undiluted over 1 to 2 minutes; higher doses (> 10 mg) to be diluted in 50 mL of compatible solution (preferably NS) and given IVPB over at least 15 minutes. **Note:** Rapid IV administration may be associated with a transient (but intense) feeling of anxiety and restlessness, followed by drowsiness.

Nasal: For intranasal administration only. Prime pump (press 10 times until spray appears) prior to first use or if spray unused ≥ 2 weeks. Insert applicator into nostril, tilt head slightly forward keeping bottle upright, and close off the other nostril. Breathe in through nose. While inhaling, press pump to release spray; exhale through mouth. Avoid spraying directly into nasal septum, eyes, or mouth. After each use, wipe the spray tip with a clean tissue and replace cap. If the spray nozzle gets clogged, clean by removing nozzle and soaking in warm water; do not insert a pin or other sharp object into the nozzle. Discard after 4 weeks, even if bottle is not completely empty.

Tablets and oral solution: For scheduled dosing, administer 30 minutes prior to meals and at bedtime.

Bariatric surgery: Oral solutions may contain nonabsorbable sugars (eg, sorbitol, mannitol, xylitol) that can cause dumping syndrome (Ref). Refer to product labeling. Providers should determine if the condition being treated can be safely monitored or if a switch to an alternative formulation or medication is necessary.

Orally disintegrating tablets: For scheduled dosing, administer on an empty stomach at least 30 minutes prior to food and at bedtime (do not repeat if inadvertently taken with food). Do not remove from packaging until time of administration. If tablet breaks or crumbles while handling, discard and remove new tablet. Using dry hands, place tablet on tongue and allow to dissolve (disintegrates

within ~1 minute [range: 10 seconds to 14 minutes]). Swallow with saliva (formulation is designed to be taken without liquids).

SUBQ administration (off-label route) has been reported, either as an intermittent bolus injection or as continuous infusion (Ref).

Preparation for Administration: Adult

Injection: Lower doses (≤ 10 mg): No dilution required; Higher doses (> 10 mg): Dilute in 50 mL of compatible solution (preferably NS).

Administration: Pediatric

Oral: Oral solution, tablet: Administer 30 minutes before meals and at bedtime.

Parenteral:

IM: May be administered IM.

IV: **Note:** Rapid IV administration is associated with a transient but intense feeling of anxiety and restlessness, followed by drowsiness.

IV push: For doses ≤ 10 mg may be administered undiluted (5 mg/mL) by direct IV push over 1 to 2 minutes.

Intermittent IV infusion: For higher doses (> 10 mg), dilute dose prior to use and administer over at least 15 minutes.

Preparation for Administration: Pediatric

Parenteral: Doses ≤ 10 mg: No dilution required; doses > 10 mg: Dilute in 50 mL of compatible solution (preferably NS) prior to administration.

Storage/Stability

Injection: Store prefilled syringes and intact vials at 20°C to 25°C (68°F to 77°F); protect from light. Injection is photosensitive and should be protected from light during storage; parenteral admixtures in D5W, D5 $\frac{1}{2}$ NS, NS, LR, or Ringer's injection are stable for up to 24 hours after preparation at normal light conditions or up to 48 hours if protected from light. When mixed with NS, can be stored frozen for up to 4 weeks; metoclopramide is degraded when admixed and frozen with D5W.

Nasal: Store at 20°C to 25°C (68°F to 77°F); excursion permitted to 15°C to 30°C (59°F to 86°F). Use within 4 weeks of opening bottle.

Oral solution/orally disintegrating tablet/tablet: Store at 20°C to 25°C (68°F to 77°F). Protect from light. Do not freeze oral solution. Keep orally disintegrating tablet in original packaging until just prior to use.

Use: Labeled Indications

Injection:

Chemotherapy-induced nausea and vomiting, prophylaxis: Prophylaxis of nausea and vomiting associated with emetogenic cancer chemotherapy. **Note:** Injectable metoclopramide prior to moderate- to high-emetic-risk chemotherapy is rarely indicated due to the potential for neurologic events and availability of more efficacious alternative agents.

Gastroparesis, diabetic: Relief of symptoms associated with acute and recurrent diabetic gastroparesis.

Postoperative nausea and vomiting, prophylaxis: Prophylaxis of postoperative nausea and vomiting in circumstances where nasogastric suction is undesirable.

Note: Although included as an FDA-approved use in the manufacturer's labeling for prevention of postoperative nausea and vomiting, available data to support use in this condition are limited (De Oliveira 2012; Gan 2020; Weibel 2021). Use may be considered when dopamine antagonists are indicated but other preferred agents in this class (eg, droperidol) cannot be used (Gan 2020).

Small bowel intubation (postpyloric feeding tube placement): Facilitation of small bowel intubation in adults and pediatric patients in whom the tube does not pass the pylorus with conventional maneuvers (within 10 minutes).

Note: Although FDA approved for facilitation of postpyloric feeding tube placement, available data to support use in this condition are limited. Systematic reviews have not shown an association between metoclopramide administration and successful postpyloric tube placement and overall quality of evidence is low; however, some individual studies have shown benefit (Hu 2018; Ouyang 2022; Silva 2015).

Nasal:

Gastroparesis, diabetic: Relief of symptoms associated with acute and recurrent diabetic gastroparesis in adults.

Oral:

Gastroesophageal reflux disease, refractory: Short-term (4 to 12 weeks) treatment in adults with documented symptomatic gastroesophageal reflux disease (GERD) who fail to respond to conventional therapy.

Note: Metoclopramide has not been shown to be safe and effective for symptomatic documented gastroesophageal reflux for ≥ 12 weeks (manufacturer's labeling). May use metoclopramide as an adjunctive therapy **only** if gastroparesis is confirmed. The American College of Gastroenterology (ACG) guidelines for the treatment of GERD recommend that diagnostic evaluation to confirm underlying gastroparesis be performed **prior to** considering the use of prokinetic agents (ACG [Katz 2013]). Furthermore, American Gastroenterological Association (AGA) guidelines for the treatment of GERD recommend **against** the use of metoclopramide as monotherapy or adjunctive therapy in patients with GERD (AGA [Kahrilas 2008]).

Gastroparesis, diabetic: Relief of symptoms associated with acute and recurrent diabetic gastroparesis in adults.

Use: Off-Label: Adult

Aspiration prophylaxis in patients undergoing anesthesia; Bowel obstruction (partial), malignant inoperable; Chemotherapy-induced nausea and vomiting, prophylaxis of delayed emesis (high-risk regimen); Dyspepsia, functional (refractory); Gastroparesis, nondiabetic; Hiccups; Medication-overuse headache or intractable migraine (status migrainosus); Migraine, severe, acute treatment (emergency setting); Nausea and vomiting: Advanced cancer-associated; Nausea and vomiting: Pregnancy-associated, severe or refractory; Nausea and/or vomiting: Undifferentiated or due to a variety of medical conditions associated with acute self-limiting nausea/vomiting; Nausea and vomiting: Vertigo-associated; Radiation therapy-induced nausea and vomiting (rescue therapy); Tension-type headache, acute (emergency setting)

Medication Safety Issues

Sound-alike/look-alike issues:

Metoclopramide may be confused with metOLazone, metoprolol, metroNIDAZOLE

Reglan may be confused with Megace, Regonol, Renagel, Regitine

Older Adult: High-Risk Medication:

Beers Criteria: Metoclopramide is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older (unless used for the treatment of gastroparesis with treatment duration < 12 weeks, except in rare cases) due to its potential for causing extrapyramidal effects, including tardive dyskinesia; these risks may be increased in frail older adults and with longer treatment duration (Beers Criteria [AGS 2023]).

Metoclopramide is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) with parkinsonism (O'Mahony 2023).

Pediatric patients: High-risk medication:

KIDs List: Dopamine antagonists, including metoclopramide, are identified on the Key Potentially Inappropriate Drugs in Pediatrics (KIDs) list due to risk of acute dystonic reactions (eg, oculogyric crisis, torticollis). Use should be avoided in neonates and infants, and used with caution in children ≥ 1 year and adolescents ≤ 18 years of age (strong recommendation; high quality of evidence) (PPA [McPherson 2025]).

Metabolism/Transport Effects

Substrate of CYP1A2 (Minor), CYP2D6 (Major); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Agents with Clinically Relevant Anticholinergic Effects: May decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Ajmaline: May increase serum concentrations of CYP2D6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Alcohol (Ethyl): May increase sedative effects of Metoclopramide. *Risk C: Monitor*

Anti-Parkinson Agents (Dopamine Agonist): Metoclopramide may decrease therapeutic effects of Anti-Parkinson Agents (Dopamine Agonist). *Risk X: Avoid*

Antipsychotic Agents: Metoclopramide may increase adverse/toxic effects of Antipsychotic Agents. *Risk X: Avoid*

Artemether and Lumefantrine: May increase serum concentrations of CYP2D6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Atovaquone: Metoclopramide may decrease serum concentrations of Atovaquone. Management: Consider alternatives to metoclopramide when possible; atovaquone should only be used with metoclopramide if no other antiemetics are available. *Risk D: Consider Therapy Modification*

Cabergoline: May decrease therapeutic effects of Metoclopramide. Metoclopramide may decrease therapeutic effects of Cabergoline. *Risk X: Avoid*

Cardiac Glycosides: Metoclopramide may decrease serum concentrations of Cardiac Glycosides. *Risk C: Monitor*

CNS Depressants: Metoclopramide may increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CycloSPORINE (Systemic): Metoclopramide may increase serum concentrations of CycloSPORINE (Systemic). *Risk C: Monitor*

CYP2D6 Inhibitors (Moderate): May increase serum concentrations of Metoclopramide. *Risk C: Monitor*

CYP2D6 Inhibitors (Strong): May increase serum concentrations of Metoclopramide. Management: For gastroparesis: reduce metoclopramide dose to 5mg 4 times/day and limit to 20mg/day; nasal spray not recommended. For GERD: reduce metoclopramide dose to 5mg 4 times/day or to 10mg 3 times/day and limit to 30mg/day. Monitor for toxicity when combined. *Risk D: Consider Therapy Modification*

Dapsone (Topical): May increase adverse/toxic effects of Methemoglobinemia Associated Agents. *Risk C: Monitor*

Deutetrabenazine: May increase adverse/toxic effects of Metoclopramide. Specifically, the risk for akathisia, parkinsonism, or neuroleptic malignant syndrome may be increased. *Risk C: Monitor*

DiazePAM: Metoclopramide may increase CNS depressant effects of DiazePAM. Metoclopramide may increase serum concentrations of DiazePAM. *Risk C: Monitor*

DroPERidol: May increase adverse/toxic effects of Metoclopramide. *Risk X: Avoid*

Fosfomycin: Gastrointestinal Agents (Prokinetic) may decrease serum concentrations of Fosfomycin. *Risk C: Monitor*

Levosulpiride: Benzamide Derivatives may increase adverse/toxic effects of Levosulpiride. *Risk C: Monitor*

Local Anesthetics: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Mavorixafor: May increase serum concentrations of CYP2D6 Substrates (High risk with Inhibitors). *Risk X: Avoid*

MetyroSINE: May increase adverse/toxic effects of Metoclopramide. *Risk X: Avoid*

Mivacurium: Metoclopramide may increase neuromuscular-blocking effects of Mivacurium. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: Metoclopramide may increase hypertensive effects of Monoamine Oxidase Inhibitors. *Risk X: Avoid*

Nitric Oxide: May increase adverse/toxic effects of Methemoglobinemia Associated Agents. Combinations of these agents may increase the likelihood of significant methemoglobinemia. *Risk C: Monitor*

Posaconazole: Metoclopramide may decrease serum concentrations of Posaconazole. *Risk C: Monitor*

Prilocaine: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Prilocaine. Combinations of these agents may increase the likelihood of significant methemoglobinemia. Management: Monitor for signs of methemoglobinemia when prilocaine is used in combination with other agents associated with development of methemoglobinemia. Avoid use of these agents with prilocaine/lidocaine cream in infants less than 12 months of age. *Risk C: Monitor*

Primaquine: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Primaquine. Specifically, the risk for methemoglobinemia may be increased. Management: Avoid concomitant use of primaquine and other drugs that are associated with methemoglobinemia when possible. If combined, monitor methemoglobin levels closely. *Risk D: Consider Therapy Modification*

Promethazine: Metoclopramide may increase adverse/toxic effects of Promethazine. *Risk X: Avoid*

QT-prolonging Agents (Highest Risk): QT-prolonging Agents (Indeterminate Risk - Caution) may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Quinagolide: Metoclopramide may decrease therapeutic effects of Quinagolide. *Risk C: Monitor*

Rivastigmine: Adverse/toxic effects of Rivastigmine and Metoclopramide may increase with coadministration. Specifically, the risk of extrapyramidal adverse reactions may be increased with this combination. Management: Consider alternatives when possible. Labeling states that this combination is not recommended because the risk of extrapyramidal adverse effects may be increased. If combined, monitor for evidence of extrapyramidal effects. *Risk D: Consider Therapy Modification*

Serotonergic Agents (High Risk): Metoclopramide may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Consider monitoring

for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Sirolimus (Conventional): Gastrointestinal Agents (Prokinetic) may increase serum concentrations of Sirolimus (Conventional). *Risk C: Monitor*

Sodium Nitrite: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Sodium Nitrite. Combinations of these agents may increase the likelihood of significant methemoglobinemia. *Risk C: Monitor*

Succinylcholine: Metoclopramide may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Tacrolimus (Systemic): Metoclopramide may increase serum concentrations of Tacrolimus (Systemic). Specifically, treatment of gastroparesis may increase tacrolimus concentrations. *Risk C: Monitor*

Tetrabenazine: Metoclopramide may increase adverse/toxic effects of Tetrabenazine. *Risk X: Avoid*

Thiopental: Metoclopramide may increase therapeutic effects of Thiopental. Management: Consider thiopental dose reduction when used concomitantly with metoclopramide. Monitor patient response to treatment closely if using this combination. *Risk D: Consider Therapy Modification*

Trimetazidine: Metoclopramide may increase adverse/toxic effects of Trimetazidine. Specifically, the risk of extrapyramidal symptoms may be enhanced. *Risk X: Avoid*

Triptorelin: Hyperprolactinemic Agents may decrease therapeutic effects of Triptorelin. *Risk X: Avoid*

Reproductive Considerations

Metoclopramide may increase prolactin concentrations; hyperprolactinemia may suppress hypothalamic GnRH, resulting in reduced pituitary gonadotropin secretion which may inhibit reproductive function by impairing gonadal steroidogenesis. Amenorrhea and impotence have been reported.

Pregnancy Considerations

Metoclopramide crosses the placenta and can be detected in cord blood and amniotic fluid (Arvela 1983; Bylsma-Howell 1983).

Based on available data, information related to the risk of major malformations following in utero exposure to metoclopramide is inconsistent (Ishikawa 2022; Sun 2021). Extrapyramidal symptoms or methemoglobinemia may potentially occur in the neonate.

Drug-induced acute dystonic reactions have been reported following use of metoclopramide in non-pregnant and pregnant patients; maternal CYP2D6 metabolizer status and hormonal changes that occur during pregnancy may contribute to this risk (Chua 2019).

Metoclopramide is one of the agents that may be considered for adjunctive treatment of nausea and vomiting in pregnant patients when symptoms persist following initial pharmacologic therapy. Oral or IM therapy may be given in patients who are not dehydrated; IV therapy should be used when dehydration is present (ACOG 189 2018). Metoclopramide may be used for prophylaxis of nausea and vomiting associated with cesarean delivery (ASA 2016; Smith 2011). Metoclopramide has also been used for the treatment of acute or tension headache in pregnant patients when acetaminophen is not effective (Childress 2018; Hamilton 2019).

Breastfeeding Considerations

Metoclopramide is present in breast milk.

Information related to the presence of metoclopramide in breast milk is available from multiple studies (Hansen 2005; Kauppila 1983; Lewis 1980).

- One study included mothers with treatment initiated between 3 and 9 days' postpartum (early puerperium, n = 5) or 8 and 12 weeks' postpartum (late puerperium, n = 18). All patients in the study were administered metoclopramide 10 mg three times daily for 2 weeks. The highest concentration of metoclopramide in breast milk (0.1565 mcg/mL) was noted after 4 days of treatment

in an early puerperium mother. The mean breast milk concentration in the late puerperium group was 0.0479 mcg/mL $\pm$ 0.0265 mcg/mL. Peak breast milk concentrations appeared approximately the same time as peak maternal serum concentrations, 2 to 3 hours after the dose. Metoclopramide was also detected in the serum of one breastfeeding infant (Kauppila 1983).

- Using a milk concentration of 0.1565 mcg/mL, the estimated exposure to the breastfeeding infant would be 0.023 mg/kg/day (relative infant dose ~4.6% based on a therapeutic infant dose of 0.5 mg/kg/day).
- According to the manufacturer, the decision to continue or discontinue breastfeeding during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. However, in general, breastfeeding is considered acceptable when the relative infant dose of a medication is <10% (Anderson 2016; Ito 2000).

Although not commonly reported, transient intestinal discomfort has been noted in two breastfed infants following exposure via breast milk (Zuppa 2010). Depression and headache have been reported in some lactating women; other adverse events reported in the mother are similar to those reported following metoclopramide use for labeled indications (Shen 2021).

Infants exposed to metoclopramide via breast milk should be monitored for extrapyramidal symptoms or methemoglobinemia. When treatment for maternal nausea and vomiting is needed, the WHO recommends metoclopramide be avoided due to insufficient data on long-term side effects to the infant (WHO 2002).

Metoclopramide may increase prolactin concentrations and cause galactorrhea and gynecomastia. As such, it has been evaluated as a galactagogue in women with insufficient milk supply. Based on pooled data from available studies, it is unclear if milk production is increased over placebo; use may be less effective following delivery of preterm infants (Foong 2020; Shen 2021). Case reports describe the use of metoclopramide (in combination with other therapies) to induce lactation in adoptive patients who wish to breastfeed (Cazorla-Ortiz 2020).

Due to the potential for adverse maternal events, a full evaluation for medical causes of low milk supply and nonpharmacologic measures should be considered prior to the use of medications as galactagogues. The Academy of Breastfeeding Medicine (ABM) does not recommend use of any specific galactagogue due to inconclusive data and potential adverse effects. Metoclopramide is associated with potentially serious adverse events in the mother. If used, dose and duration of therapy should be limited (ABM [Brodrick 2018]). Avoid use in patients at increased risk for depression (Shen 2021).

Monitoring Parameters

Signs of tardive dyskinesias, extrapyramidal symptoms; signs/symptoms of neuroleptic malignant syndrome; mental alertness.

Mechanism of Action

Metoclopramide blocks dopamine receptors and (when given in higher doses) also blocks serotonin receptors in chemoreceptor trigger zone of the CNS; enhances the response to acetylcholine of tissue in upper GI tract causing enhanced motility and accelerated gastric emptying without stimulating gastric, biliary, or pancreatic secretions; increases lower esophageal sphincter tone

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Oral: 30 to 60 minutes; IV: 1 to 3 minutes; IM: 10 to 15 minutes.

Duration: Therapeutic: 1 to 2 hours, regardless of route.

Absorption: Oral: Rapid, well absorbed.

Metabolism: Hepatic via oxidation and glucuronide and sulfate conjugation; forms oxidative metabolite monodeethylmetoclopramide via CYP2D6.

Distribution: V_d : Neonates, PMA 31 to 40 weeks: 6.94 L/kg (Kearns 1998); Infants: 4.4 L/kg; Children: 3 L/kg; Adults: ~3.5 L/kg.

Protein binding: ~30%.

Bioavailability: Nasal: 47%; Oral: 80% ± 15.5%.

Half-life elimination: Normal renal function: Neonates, PMA 31 to 40 weeks: 5.4 hours (Kearns 1998); Infants: 4.15 hours (range: 2.23 to 10.3 hours) (Kearns 1988); Children: ~4 hours (range: 2 to 12.5 hours); half-life and clearance may be dose-dependent; Adults: 5 to 6 hours (may be dose dependent); Nasal: ~8 hours.

Time to peak, serum: Neonates, PMA 31 to 40 weeks: 2.45 hours (Kearns 1998); Infants: 2.2 hours; Adults: 1 to 2 hours.

Excretion: Urine (~85%); feces.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: AUC increased ~2-fold in patients with moderate to severe renal impairment and ~3.5 fold in patients with end-stage renal disease on dialysis.

Hepatic function impairment: Clearance was reduced by ~50% and half-life prolonged by ~2-fold in patients with severe hepatic impairment (Albani 1991).

CYP2D6 poor metabolizers: Elimination may be slowed.

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and as follows, must be dispensed with this medication:

Gimoti: https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/209388s001lbl.pdf#page=17

Metoclopramide oral solution: <https://www.fda.gov/media/80163/download>

Metozolv ODT: https://www.accessdata.fda.gov/drugsatfda_docs/label/2019/022246s009lbl.pdf#page=18

Reglan injection: https://www.accessdata.fda.gov/drugsatfda_docs/label/2026/017862s64lbl.pdf#page=17

Reglan ODT: https://www.accessdata.fda.gov/drugsatfda_docs/label/2011/021793s011mg.pdf

Reglan tablet: <https://www.fda.gov/media/79804/download>

Patient Counseling Points

(For additional information see "Metoclopramide: Patient drug information")

What is this drug used for?

- It is used to treat gastroesophageal reflux disease (GERD; acid reflux).
- It is used to relieve the symptoms of slow stomach emptying (gastroparesis) in people with diabetes.
- It is used to prevent upset stomach and throwing up after chemo or surgery.
- It may be given to you for other reasons. Talk with the doctor.

Children:

- This drug is not approved for use in children. However, the doctor may decide the benefits of taking this drug outweigh the risks. If your child has been given this drug, ask the doctor for information about the benefits and risks. Talk with the doctor if you have questions about giving this drug to your child.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

- Feeling dizzy, sleepy, tired, or weak
- Diarrhea, upset stomach, or throwing up

- Headache

Nose spray:

- Change in taste

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Tardive dyskinesia like trouble controlling body movements or problems with your tongue, face, mouth, or jaw like tongue sticking out, puffing cheeks, mouth puckering, or chewing
- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Shakiness, trouble moving around, or stiffness
- Restlessness
- Trouble sleeping
- Trouble controlling body movements, twitching, change in balance, trouble swallowing or speaking
- Shortness of breath, a big weight gain, or swelling in the arms or legs
- Fast, slow, or abnormal heartbeat
- Change in how often urine is passed
- Not able to control passing urine
- Change in eyesight
- Fever, chills, or sore throat
- Enlarged breasts, nipple discharge, not able to get or keep an erection, or menstrual changes
- Neuroleptic malignant syndrome like fever, muscle cramps or stiffness, dizziness, very bad headache, confusion, change in thinking, fast heartbeat, heartbeat that does not feel normal, or are sweating a lot
- Serotonin syndrome like agitation; change in balance; confusion; hallucinations; fever; fast or abnormal heartbeat; flushing; muscle twitching or stiffness; seizures; shivering or shaking; sweating a lot; severe diarrhea, upset stomach, or throwing up; or severe headache
- Depression, suicidal thoughts, emotional ups and downs, abnormal thinking, anxiety, or lack of interest in life
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Gastrobid | Meclopran | Paspertin | Plasil | Premosan | Primperan | Pylomid;
- (AR) Argentina: Fada metoclopramida/celit | Fonderyl | Lizarona | Metoclopramida | Metoclopramida gobbi | Metoclopramida hlb | Metoclopramida larjan | Metoclopramida lavimar | Metoclopramida vannier | Midatenk | Novomit | Pepsipancreol | Primavera-n | Primaveran | Primperil | Reliveran | Rilaquin | Rupemet | Trimpol;
- (AT) Austria: Gastrosil | Metoclopramidhydrochlorid Accord | Metogastron | Paspertin;
- (AU) Australia: Apo-metoclopramiæ | Emexlon | Maxolon | Metoclopramide | Metoclopramiæ actavis | Metoclopramiæ an | Metoclopramide rbx | Pramin;
- (BD) Bangladesh: Antimet | Flatelon | Flatilon | Maxil | Maxocol | Meclid | Mepralon | Metocil | Metocol | Metomid | Motilon | Movlan | Nutramid;
- (BE) Belgium: Docmetoclo | Metoclopramide efeka | Mobistal | Primperan;
- (BF) Burkina Faso: Pradis;
- (BG) Bulgaria: Cerucal | Degan | Paspertin | Reglan;
- (BR) Brazil: Cloridrato de metoclopramida | Emetic | Enzilom | Eucil | Furp metoclopramid | Furp metoclopramida | Hyposil | Istasil | Metoclosan | Metoplammin | Nausil | Noprosil | Novosil | Plabel | Plagex | Plamidasil | Plamivon | Plasil | Pramil | Remetin | Vomistop | Vonil;
- (BZ) Belize: Metoclopramida;
- (CH) Switzerland: Gastrosil | Paspertin | Primperan;
- (CL) Chile: Hemibe | Itan | Metoclopramida | Peryan;
- (CN) China: Metoclopramiæ;
- (CO) Colombia: Bonadoxina | Bonaigest | Carnotprim | Digespar | Enzimar | Enzimar nf | Metoclopramida | Metroprasil | Novomit | Plasil | Primperan;
- (CR) Costa Rica: Dibotin | Metoclopramida | Metoril;
- (CZ) Czech Republic: Cerucal | Degan | Mcp Hexal | Paspertin | Primperan | Reglan;
- (DE) Germany: Cerucal | Dura mcp | Gastronerton | Gastrotranquil | Maltyl | Mcp 1a pharma | Mcp abz | Mcp al | Mcp heumann | Mcp Hexal | Mcp ratiopharm | Mcp stada | Mcpham | Metoclopramid | Paspertin;
- (DO) Dominican Republic: Clo-Prim | Clopafar | Clopramin | Metagliz | Metoclopramida | Pramin | Primperan | Propace;
- (EC) Ecuador: Clodopan | Clopan | Clopral | Metoclopramida | Metoclopramida nifa | Metoclox | Primperan;
- (EE) Estonia: Cerucal | Cerucal-v | Mcp ratiopharm | Metoclopramid | Metoclopramide | Metoclopramide Accord;
- (EG) Egypt: Clopram | Meclopram | Metoclopramide | Migaura | Plasil | Plemazol | Primperan;
- (ES) Spain: Metoclopramida accord | Metoclopramida pensa | Primperan;
- (ET) Ethiopia: Metoclopramiæ | Premosan | Regurg;
- (FI) Finland: Emperal | Metoclopramide Accord | Metoclopramide orion | Metopram | Primperan;
- (FR) France: Anausin | Anausin Metoclopramiæ | Metoclopramide | Metoclopramide Dci | Metoclopramide gnr | Primperan | Sorbiperan;
- (GB) United Kingdom: Gastrese | Gastrobid | Maxolon | Metoclopramide | Metoclopramiæ | Metoclopramide cox | Metoclopramiæ dc | Metox | Metramid | Mygdalon | Parmid | Primperan;
- (GR) Greece: Primperan;
- (HK) Hong Kong: Apo-Metoclop | Bf-Metoclopramide | Clopamon | Cloperan | Clopramel | Hawkperan | Ma | Maril | Martomide | Maxolon | Mecloma | Metoclopramiæ | Metolon | Metomide | Metram | Primperan | Prowel | Pulin | Sinprim | Syntomide;
- (HR) Croatia: Metopran | Reglan;
- (HU) Hungary: Cerucal | Paspertin;
- (ID) Indonesia: Clopramel | Damaben | Emeran | Enakur | Ethiferan | Gavistal | Hufaclop | Lexapram | Mepramide | Metoclopramiæ | Metolon | Navoren | Nilatika | Nofoklam | Noristal

- | Normastin | Omevomid | Opram | Piralen | Plasil | Praminal | Primperan | Raclonid | Reguloop | Sanmeto | Sotatic | Tomit | Topram | Vertivom | Vilapon | Vomiles | Vomipram | Vomitol | Vosea | Zumatrol;
- (IE) Ireland: Antimet | Gastrobid | Maxolon | Metocyl | Primperan;
- (IL) Israel: Pramin;
- (IN) India: Almeto | Emenil | Emenorm | Maxeron | Maxinorm | Metanorm | Metochlopramide | Metoclopramide | Metocontin | Metonorm | Metotid | Perinorm | Promet | Reggi | Reglan | Sigmet | Vominorm;
- (IQ) Iraq: Meclostin | Meclokindin | Vomisaf;
- (IT) Italy: Metoclopramide Accord | Plasil | Pramidin;
- (JO) Jordan: Clopram | Damperide | Elitan | Gastroprid | Plasil | Primperan | Pylomid;
- (JP) Japan: Anolexinon | Donopon gp | Elieten | Emperan | Folicron | Gaputazis | Gosper m | Metoclo | Metoclopramide | Metoclopramide hydrochloride takata | Metoclopramide takata | Metoclopramide towa | Metoclopramide tsuruhara | Moriperan | Peraprin | Pramiel | Primperan | Prinparl | Prometin | Putoprin | Terperan;
- (KE) Kenya: Asmeto | Cloperan | Emeton | Gastrolon | Melasil | Metcos | Primperan;
- (KR) Korea, Republic of: Gasrobi | Macperan | Metocron | Mexolon | Prometin | Sinil metoclopramide;
- (KW) Kuwait: Gastrobid | Maxolon | Metocyl | Premosan | Primperan | Pylomid;
- (LB) Lebanon: Gastropride | Primperan;
- (LT) Lithuania: Apo metoclop | Cerucal | Degan | Emperal | Metamol | Metoclopramide | Metoclopramide bp | Nevomitan | Paspertin | Reglan;
- (LU) Luxembourg: Gastrosil | Mcp Hexal | Mcp ratiopharm | Movistal | Primperan;
- (LV) Latvia: Apo metoclop | Cerucal | Degan | Emperal | Metoclopramide | Paspertin | Reglan;
- (MA) Morocco: Cloprame | Metagliz | Primperan | Sorbiperan | Vomistop;
- (MX) Mexico: Biopram | Camiprida | Carnotprim | Carnotprim lp | Cirulan | Clopramid | Conysmin | Dirpasid | Dolmisin | Eslinmor | Eudiges | Gigemet | Goyodem | Hopram | Meclomid | Metoclopram biomep | Metoclopram bruluart | Metoclopram gi | Metoclopram gi ly | Metoclopram provacs | Metoclopram Ultra | Metoclopram wyeth | Metoclopramida | Metoclopramida Loeffler | Metoclopramida Novartis | Metoclopramida Proquigama | Metoclopramida Raam | Metoril | Mipramid | Plasil | Polcotec | Pradex | Pramilem | Premier | Primperan | Propace | Synespramid | Tinolpro;
- (MY) Malaysia: Anpro Metoclopramide | Antimet | Axcel Metoclopramide | Emecon | Emeliv | Gastrolon | Maril | Maxolon | Metoclopramide | Metocyl | Metolon | Metomide | Mexomide | Primperan | Pulin | Setromide;
- (NL) Netherlands: Metoclopramide HCL | Metoclopramide hcl Alharma | Metoclopramide Hcl CF | Metoclopramide Hcl PCH | Metoclopramide hcl ratiopharm | Metoclopramidemonohydrochloride accord | Primperan;
- (NO) Norway: Afipran | Mcp al | Metoclopramide Accord | Metoclopramide euromedica | Metoclopramide nmd grossist | Metoclopramide orifarm | Primperan;
- (NZ) New Zealand: Maxolon | Metamide | Metoclopramide | Metoclopramide actavis | Metoclopramide baxter;
- (PA) Panama: Clo-Prim | Clopram | Clopramida;
- (PE) Peru: Aeronase | Apo-metoclopramida | Asinpramida | Cloromel | Fluxt | Hemesys | Metoclopramida | Primperan;
- (PH) Philippines: Meto | Metoclopramide | Myclosil | Pasclomet | Paspertin | Plasil | Rafromide | Reglan;
- (PK) Pakistan: Clopan | Cloprel | Gastrolon | Gastrusil | Kanamide | Maxaclor | Maxolon | Mecomide | Mediclop | Metalon | Metoclon | Metoclop | Metoclopramide | Metomide | Plasil | Regelan | Stomac | Unexolon | Vomilide;
- (PL) Poland: Cerucal | Degan | Metoclopramide | Metoclopramidum polpharma | Passpertin;
- (PR) Puerto Rico: Metoclopramide | Metoclopramide HCL | Reglan;
- (PT) Portugal: Mecloprina | Metgen | Metoclopramida | Primperan;
- (PY) Paraguay: Digesplen | Emelit | Glicimet | Imperan | Reliveran;
- (QA) Qatar: Apo-Metoclop | Cloperan | Clopram | Metonex | Metoram | Metpamid | Premosan (Metoclopramide) | Primperan (Metoclopramide HCl) | Vomepram | Vominorm;
- (RO) Romania: Cerucal | Degan | Metoclopramid;

- (RU) Russian Federation: Apo metoclop | Cerucal | Ceruglan | Cerulan | Melomid | Metoclopramiæ | Metoclopramiæ velfarm | Metoclopramide-akr | Perinorm | Reglan | Vero metoclopramid;
- (SA) Saudi Arabia: Metosil | Plasil | Premosan | Primperan | Pylomid;
- (SE) Sweden: Metoclopramiæ Accord | Metoclopramide orifarm | Metoclopramiæ orion | Metoklopramid alternova | Primperan;
- (SG) Singapore: Apo-Metoclop | Maril | Maxolon | Mecloma | Metoclopramide | Metocyl | Metolon | Primperan | Syntomide;
- (SI) Slovenia: Reglan;
- (SK) Slovakia: Cerucal | Degan | Paspertin;
- (SL) Sierra Leone: Metoclop | Metoclopramiæ;
- (SV) El Salvador: Meclopram;
- (TH) Thailand: Elitan | Emetal | Gensil | Hawkperan | K.b.meta | Manosil | Maril | Maxolon | Mepamide | Meramide | Metoclopramide | Metoclor | Metono | Nausa | Nausil | Nausin | Noosea | Plamide | Plamine | Plasil | R J | Seapcosil | Tomed | Vasil | Vomesea | Vomitin | Vosil;
- (TN) Tunisia: Mepramiæ | Pramidyl per os | Primperan;
- (TR) Turkey: Metoklamid | Metpamid | Primperan;
- (TW) Taiwan: Aswell | Balon | Biweslan | Chiaowelgen | Chitou | Dringen | Eline | Enteran | Imperam | Linperan | Meniperan | Mepram | Mepramide | Methoclopramide | Metoclopramide | Metoco | Metopelan | Metoperan | Metoperon | Mevaperan | Pantomide | Paspertin | Perone | Plasil | Plindan | Polun | Poriran | Prenperon | Prevomic | Primlan | Primperan | Primram | Primran | Prinparl | Pripram | Promeran | Prometin | Prowel | Pulin | Pulinpelin | Pulperan | Pusuan | Putelome | Rotelan | Rowelcon | Sinprim | Sinthato | Sinvomin | Suvecon | Suweilan | Torowilon | Wei Lian | Weiperan | Winperan | Zudaw | Zuperan;
- (UA) Ukraine: Cerucal | Cerulan | Metamol | Metoclopramid | Metoclopramiæ | Metucal | Perinorm;
- (UG) Uganda: Asmeto | Camet | Metapromide | Meto | Metoclopramiæ | Metoprix | Prasil;
- (UY) Uruguay: Caillomida | Comportan | Legir | Metoclopramida | Novomit | Primperan | Revalidan;
- (VE) Venezuela, Bolivarian Republic of: Clop | IrtoPan | Mepramida | Metoclopramida | Pradamin | Pramide | Primperan;
- (VN) Viet Nam: Kanausin | Meclopstad | Perimirane | Pimeran;
- (ZA) South Africa: Ametic | Betaclopramide | Bio metoclopramiæ | Clopamon | Contromet | Maxolon | Merck-metoclopramide | Rolab-metoclopramiæ | Setin;
- (ZM) Zambia: Metamide | Metoclopramide | Regurg;
- (ZW) Zimbabwe: Clopamon | Metoclopramiæ

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